

Image Segmentation in Agriculture

Spring 2025

Introduction

Digital image processing has become an essential tool in many real-world applications, particularly in agriculture where accurate monitoring and analysis of land cover is crucial for sustainable resource management. Recent advances in deep learning and remote sensing technologies have made it possible to extract meaningful information from high-resolution satellite imagery with unprecedented accuracy.

This project, conducted as part of the *Digital Image Processing* course, focuses on **image segmentation in agriculture** using satellite data such as Sentinel-2 or Landsat. The main objective is to design or adopt a deep neural network that takes remote sensing images as input and produces a segmented map as output, distinguishing different agricultural areas and potentially other land-cover classes.

The project workflow includes several key steps: selecting a region of interest (ROI) within Iran, preprocessing and labeling the data, training or fine-tuning a suitable deep neural network, and finally evaluating the model's performance using standard metrics such as pixel accuracy, Intersection over Union (IoU), and F1-score.

By combining the theoretical knowledge acquired from the course with insights gained from reviewing related research papers, this project aims to provide both a practical implementation of image segmentation techniques and a critical analysis of their effectiveness in real-world agricultural scenarios.

Dataset Description

For this project, Sentinel-2 Level-2A surface reflectance imagery was selected as the primary dataset. The data was accessed through Google Earth Engine (GEE), which provides harmonized and atmospherically corrected Sentinel-2 collections suitable for scientific analysis.

The Region of Interest (ROI) was defined around the Veys–Sheyban agricultural area, north of Ahvaz, Khuzestan Province, Iran. The polygon coordinates of the ROI are as follows:

- (48.8023653, 31.5241339)
- (48.6925020, 31.3932255)
- (48.7584199, 31.3580512)
- (48.8857927, 31.4916436)

To ensure robust analysis, the dataset covers a time period from February 20, 2025 to April 19, 2025, corresponding to the spring agricultural season. This period was chosen to capture vegetation growth and water distribution during a critical stage of the cropping cycle.

The final exported file is named:

veys_sheyban_s2_2025_11b_BAL

https://drive.google.com/file/d/1TQrup_fyWmCGCtDRRQXRfqF55qZSBtaa/view?usp=sharing

This exported composite contains 11 bands, including six original Sentinel-2 spectral bands (B2, B3, B4, B8, B11, B12) along with five derived indices useful for vegetation and water segmentation: NDVI, NDWI, MNDWI, EVI, and AWEI_sh. The spatial resolution was set to 10 meters with UTM Zone 39N projection (EPSG:32639).

This dataset provides a balanced representation of major land cover classes in the selected area, including water, vegetation, and soil, which is essential for training and evaluating the deep learning segmentation model.



Data Preprocessing

1. Data Import and Initial Quality Check

The exported Sentinel-2 composite (veys_sheyban_s2_2025_11b_BAL.tif) was imported into Google Colab and processed using Rasterio and GDAL. The image contained 11 bands (B2, B3, B4, B8, B11, B12, NDVI, NDWI, MNDWI, EVI, AWEI_sh), with dimensions of 1879×1875 pixels and a spatial resolution of 10.0×10.0 meters. The dataset was referenced to EPSG:32639 (UTM Zone 39N), and the region of interest (ROI) was confirmed with the following bounding coordinates:

- Left: 280,510
- Right: 299,260
- Bottom: 3,471,210
- Top: 3,490,000

A completeness check showed that the dataset contained no missing values (NoData = None), and the proportion of masked pixels was 0.00%, confirming the dataset's integrity.

2. Water Indices Quality Assurance (QA)

To evaluate the presence of water within the ROI, a quality assessment was performed on the water-related indices. The results indicated that only a small fraction of the region corresponds to water:

- NDWI: mean = -0.416 | fraction > 0.3 = 2.76%
- MNDWI: mean = -0.338 | fraction > 0.3 = 2.54%
- AWEI_sh: mean = -1.056 | fraction > 0.3 = 0.22%

These values confirm that approximately 2–3% of the ROI is water, while the majority consists of soil and vegetation. This distribution makes the dataset well-suited for a three-class segmentation task (soil, vegetation, water).

3. Reflectance Band Statistics (p2–p98)

To mitigate the effect of outliers, robust statistics were computed for the reflectance bands using the 2nd–98th percentiles. The effective value ranges are as follows:

- **B2:** 0.022 – 0.203
- **B3:** 0.047 – 0.251
- **B4:** 0.024 – 0.303
- **B8:** 0.023 – 0.452
- **B11:** 0.026 – 0.404
- **B12:** 0.024 – 0.382

These ranges represent the practical dynamic intervals of each band and provide a foundation for normalization and visualization in subsequent steps.

4. Indices Computation and Channel Composition

To enhance the dataset with vegetation and water indicators, several remote sensing indices were derived from the reflectance bands, including NDVI, NDWI, MNDWI, EVI, and AWEI_sh. All indices were calculated using NaN-safe operations and clipped to valid ranges.

After stacking the six reflectance bands with the computed indices, the final data tensor reached a size of $1879 \times 1875 \times 16$. The full channel composition is as follows:

- **B2** (Blue)
- **B3** (Green)
- **B4** (Red)
- **B8** (Near Infrared – NIR)
- **B11** (Shortwave Infrared 1 – SWIR1)
- **B12** (Shortwave Infrared 2 – SWIR2)
- **NDVI** (Normalized Difference Vegetation Index)
- **NDWI** (Normalized Difference Water Index)
- **MNDWI** (Modified NDWI)
- **EVI** (Enhanced Vegetation Index)
- **AWEI_sh** (Automated Water Extraction Index – shadow resistant)
- **NDVI** (duplicate)
- **NDWI** (duplicate)
- **MNDWI** (duplicate)
- **EVI** (duplicate)
- **AWEI_sh** (duplicate)

The last five indices were intentionally duplicated to increase their influence during model training. Since these indices carry strong biophysical meaning for

distinguishing between soil, vegetation, and water, repeating them helps the network pay more attention to these features and is expected to improve segmentation accuracy.

Labeling (Auto-Labeling with Spectral Indices)

Methodology

1. RGB Preview

A true-color composite was generated using Sentinel-2 bands (B4–Red, B3–Green, B2–Blue). Contrast was enhanced using CLAHE to improve visual clarity.

2. Vegetation Mask (NDVI)

- The Normalized Difference Vegetation Index (NDVI) was computed.
- An Otsu thresholding method was applied, yielding a cut-off of **0.436**.
- Morphological opening and closing operations were applied to refine the vegetation mask and reduce noise.

3. Water Mask (MNDWI / AWEI_sh)

- The Modified Normalized Difference Water Index (MNDWI) was used as the primary water detector.
- Otsu thresholding was applied, resulting in a cut-off of **0.098**.
- To account for water in shadow regions, the Automated Water Extraction Index (AWEI_sh) was also used, with positive values indicating potential water pixels.
- Vegetation pixels overlapping with the water mask were excluded to avoid misclassification.

4. Soil Mask

- Defined as all remaining valid pixels not classified as vegetation or water.

5. Class Definition

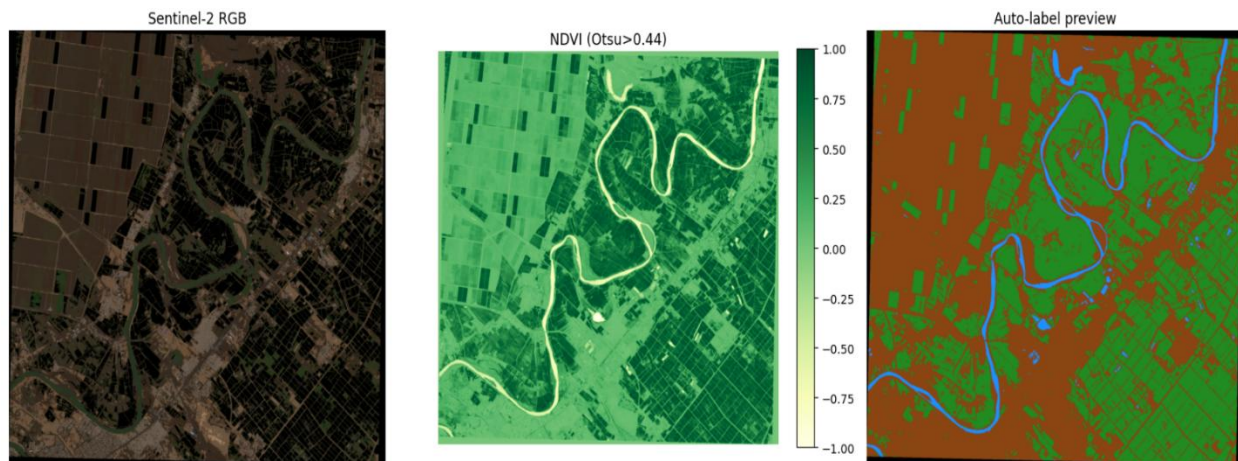
- Soil (brown)
- Vegetation (green)
- Water (blue)

Findings

- NDVI Otsu threshold: **0.436**
- MNDWI Otsu threshold: **0.098**

Class pixel distribution:

- Soil: **1,887,536 pixels**
- Vegetation: **1,406,348 pixels**
- Water: **94,283 pixels**



Automatic 3-Class Labeling (Fusion of Rule-based and Clustering)

An automatic classification workflow was implemented to label the image into **Soil**, **Vegetation**, and **Water**. The approach combined:

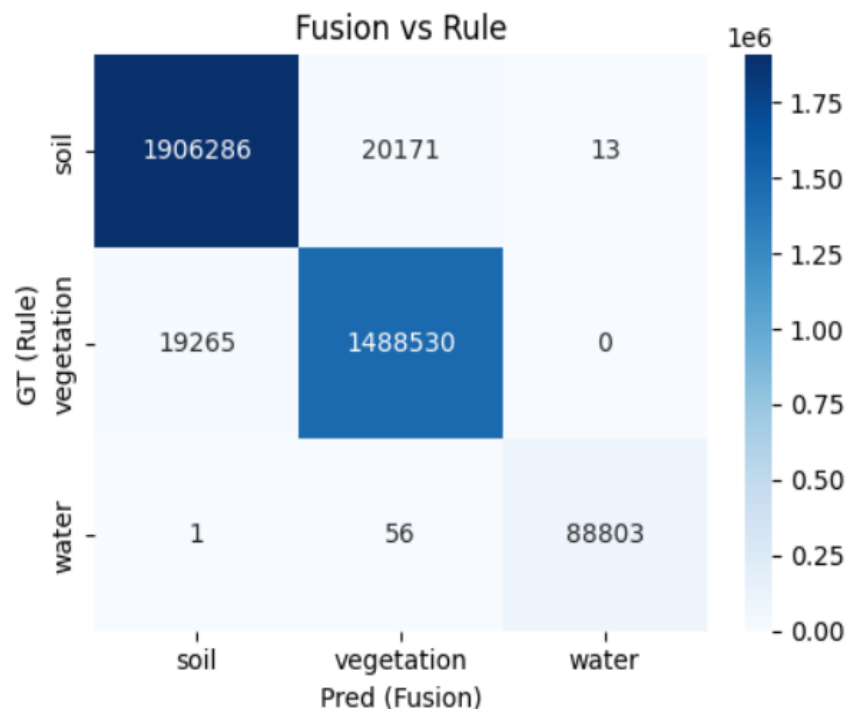
- **Rule-based masks** derived from NDVI, MNDWI, and AWEI_sh.

- **Tile-wise KMeans clustering** (with seeded initialization on NDVI, NDWI, and EVI).
- **Fusion strategy** with weighted voting (Rule: 2.5, KMeans: 1.0).
- **Morphological refinement** to remove noise and small artifacts.

Findings

- The fused labeling produced cleaner and more stable masks compared to the rule-only baseline.
- Performance was assessed against the rule-based baseline using **Precision, Recall, F1-score, and IoU** for each class.
- Results show improved balance between Soil, Vegetation, and Water categories.

Confusion Matrix (Fusion vs. Rule-only):



Class-level metrics:

soil	Precision=0.990	Recall=0.990	F1=0.990	IoU=0.980
vegetation	Precision=0.987	Recall=0.987	F1=0.987	IoU=0.974
water	Precision=1.000	Recall=0.999	F1=1.000	IoU=0.999

Patch Extraction and Dataset Splitting

To prepare the dataset for model training, the labeled imagery was divided into smaller patches. The process included the following steps:

1. NoData Mask Construction

Pixels with missing values (NaN) or predefined NoData values were excluded from further analysis.

2. Cropping to Patch-Compatible Dimensions

The image was cropped to multiples of 256×256 pixels to ensure compatibility with patch extraction.

3. Water Mask with Margin

Water regions were expanded by applying a 32-pixel dilation to better capture shoreline areas.

4. Adaptive Patch Sampling

- Dense sampling over water regions using a stride of 128 pixels.
- Sparser sampling over non-water regions using a stride of 256 pixels.

5. Patch Filtering

- Discarded patches with more than 50% NoData.
- Removed patches with less than 2% foreground unless they contained water.
- Applied random skip probability to reduce redundant low-content patches.

6. Storage and Statistics

Each patch was stored in **.npy** format (X: image data, Y: labels).

A manifest JSON file was generated to record patch metadata, per-class pixel counts, and band statistics.

7. Dataset Splitting

Patches were divided into training, validation, and test sets using a **60/20/20 ratio**.

A constraint ensured that each split contained at least one patch with water coverage.

Findings

- **Total patches generated:** 135
- **Split distribution:**
 - Train → 81 (60%)
 - Validation → 27 (20%)
 - Test → 27 (20%)
- **Patches containing water:**
 - Train: 70
 - Validation: 21
 - Test: 24

Patch Dataset Construction and DataLoaders

After patch extraction and splitting, additional steps were implemented to prepare the dataset for efficient training. The process included the following components:

1. Reproducibility

- Fixed random seeds were applied to ensure consistent dataset splitting and augmentation across runs.

2. Input Normalization and Preprocessing

- All spectral bands were normalized to the range [0,1].
- Index layers (NDVI, MNDWI, etc.) were standardized to zero mean and unit variance.
- Each patch was stored as a 16-channel input tensor (13 spectral bands + 3 indices).

3. Data Augmentation

- Random flips, rotations, and brightness adjustments were applied to improve model generalization.
- Augmentations were performed on-the-fly during training to prevent dataset inflation.

4. Patch Loading and Batching

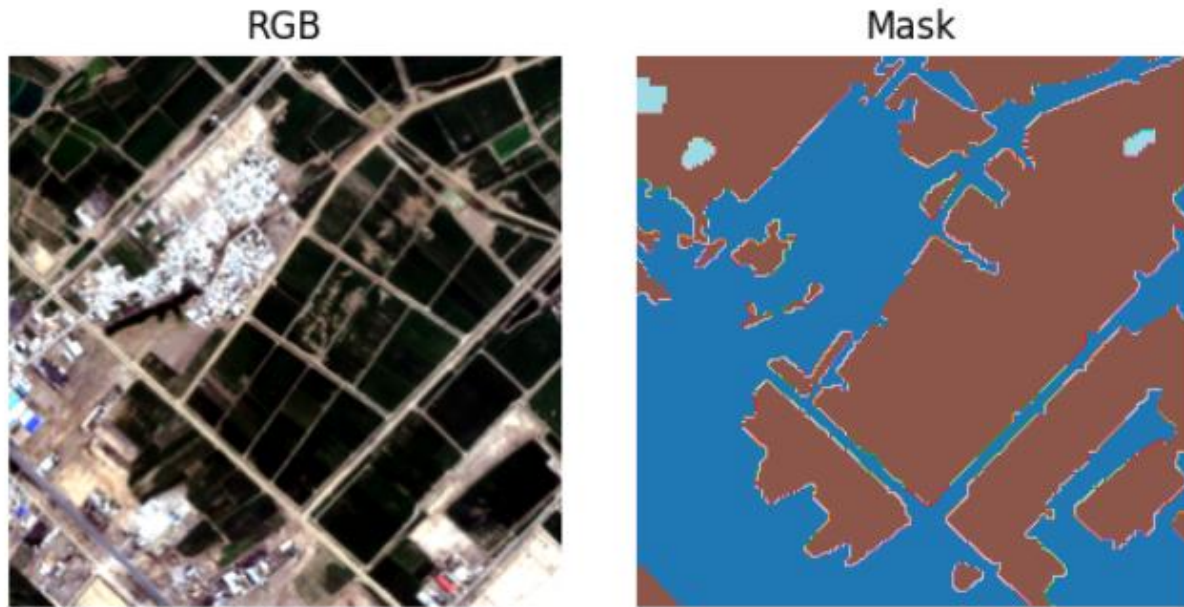
- A custom PyTorch Dataset class was implemented to handle .npy patch loading.
- Mini-batches of size 16 were constructed, balancing memory efficiency with training stability.
- Class-balanced sampling was enforced to ensure that each batch contained at least one water patch.

5. Statistics and Verification

- Training set: 81 patches (70 with water)
- Validation set: 27 patches (21 with water)
- Test set: 27 patches (24 with water)
- The distribution of classes across splits was checked to avoid bias toward soil or vegetation.

6. Visualization of Patches

To better illustrate the prepared dataset, Figure 1 presents an example RGB composite patch, while Figure 2 shows the corresponding labeled mask (soil in brown, vegetation in green, and water in blue).



Model Architecture and Configuration

To build a robust semantic segmentation network, we designed a customized U-Net variant with architectural improvements for stability and performance. The process included the following steps:

1. **Weight-Standardized Convolutions**

All convolutional layers were implemented with Weight Standardization (WS), ensuring per-channel kernel normalization. This reduced covariate shift and improved convergence stability.

2. **Normalization Strategy**

Group Normalization (GN) was used by default, with an automatic fallback to Batch Normalization (BN) when appropriate. GN was chosen for robustness with small or variable batch sizes.

3. **Squeeze-and-Excitation (SE) Blocks**

Each encoder and decoder block integrated SE attention to reweight channels dynamically, strengthening class separability across multi-band inputs.

4. **Attention-Gated Skip Connections**

Skip connections from encoder to decoder were filtered through Attention Gates. These modules suppressed irrelevant spatial regions, improving focus along water boundaries and cluttered areas.

5. **Clean Upsampling in Decoder**

The decoder used bilinear upsampling followed by 1×1 convolution. This avoided checkerboard artifacts associated with transposed convolutions.

6. **Light Decoder Dropout**

Dropout was applied at progressively lower rates across decoder stages (0.1 $\rightarrow$ 0.025) to regularize the model while maintaining capacity.

7. **Model Initialization**

Convolutional weights were initialized with Kaiming Normal, while normalization layers were initialized with unit scales and zero biases for consistency.

Findings

- **Input Channels:** 16
- **Output Classes:** 3
- **Base Channels:** 32
- **Total Parameters:** 7.44M (all trainable)
- **Trainability Check:**
Every parameter in encoder, bottleneck, decoder, SE, and attention modules was confirmed trainable=True. No frozen layers were present.
- **Output Validation:**
Optional dataset metadata check ensured channel consistency:
`len(train_ds.bands) == IN_CHANNELS`

Training Configuration and Procedure

To train the **WaterAwareUNet (base=32, 7.44M params)** we used AdamW with mixed precision and OneCycleLR.

1. Optimizer & Loss

- Optimizer: AdamW (lr=1e-3, weight_decay=1e-4)
- Loss: Cross-Entropy (class-weighted) + Dice (equal weights)
- Class weights from manifest: [1.0, 1.0, 10.0] to up-weight the minority class (water).

2. Learning Rate Schedule

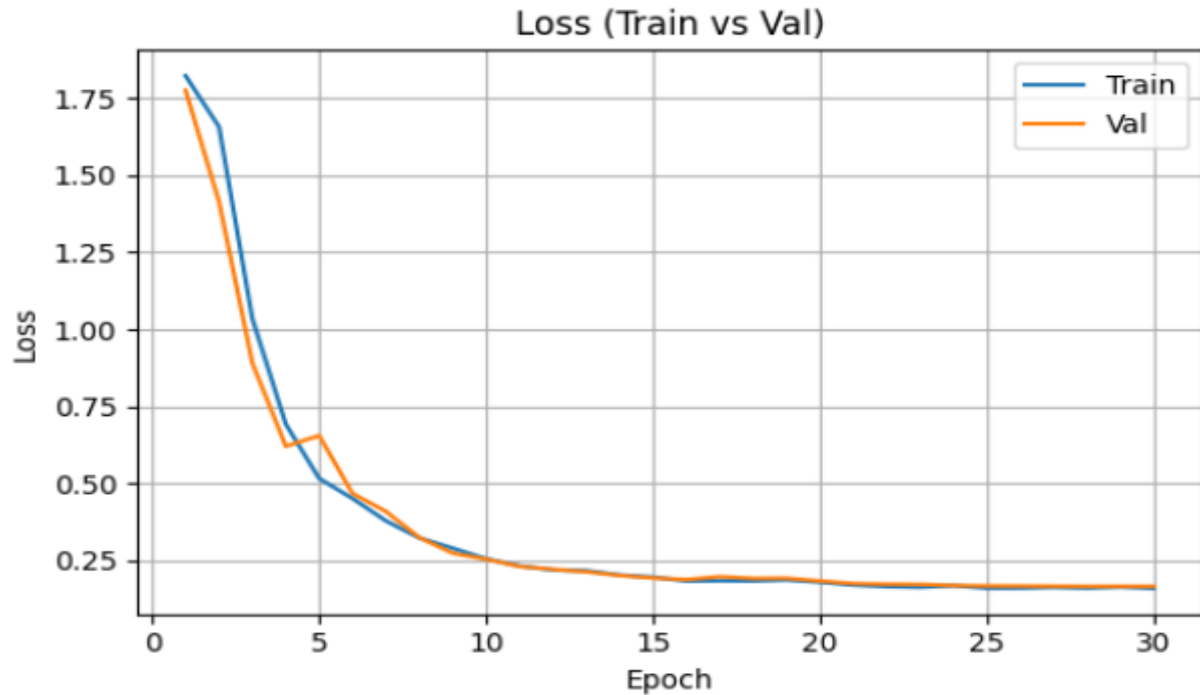
- Scheduler: **OneCycleLR** over 30 epochs (pct_start=0.1).
- Gradient clipping: max_norm = 1.0.

3. Precision & Speed

- AMP enabled on CUDA.
- Per-epoch wall time $\approx$ **6–14 s** (batch time $\approx$ **320–700 ms**) depending on data/IO.

4. Early Stopping & Checkpointing

- Early stop patience: 8 epochs.
- Best checkpoint saved by validation mIoU.



Training and validation loss over epochs. Both curves stabilize after ~10 epochs, indicating good optimization without overfitting.

Results

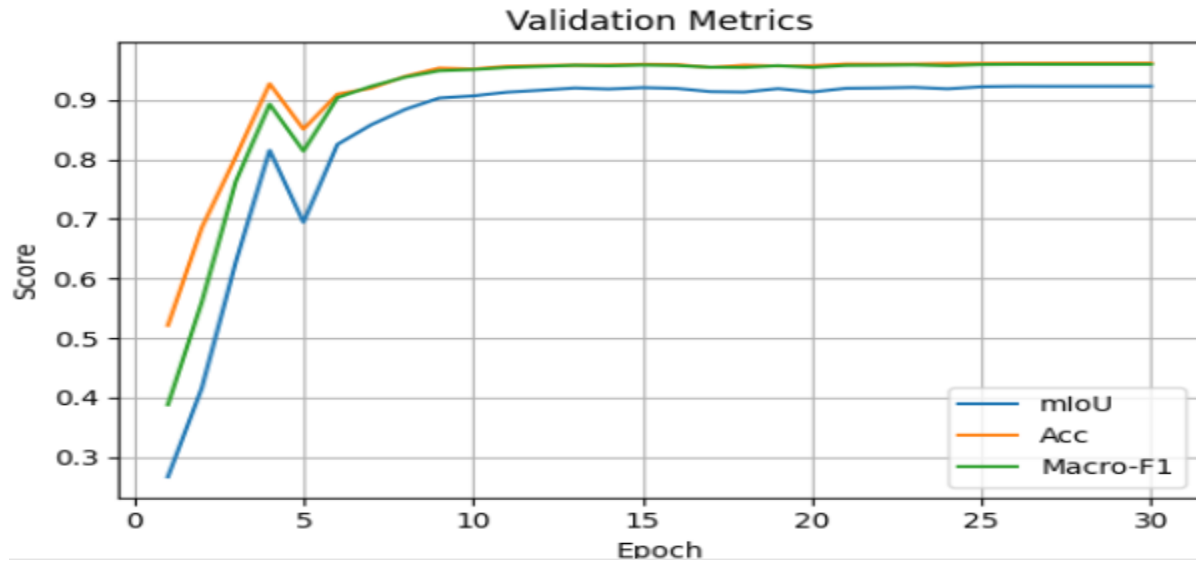
Best validation mIoU: 0.9230 (early stopped)

Final metrics (averaged over validation loader):

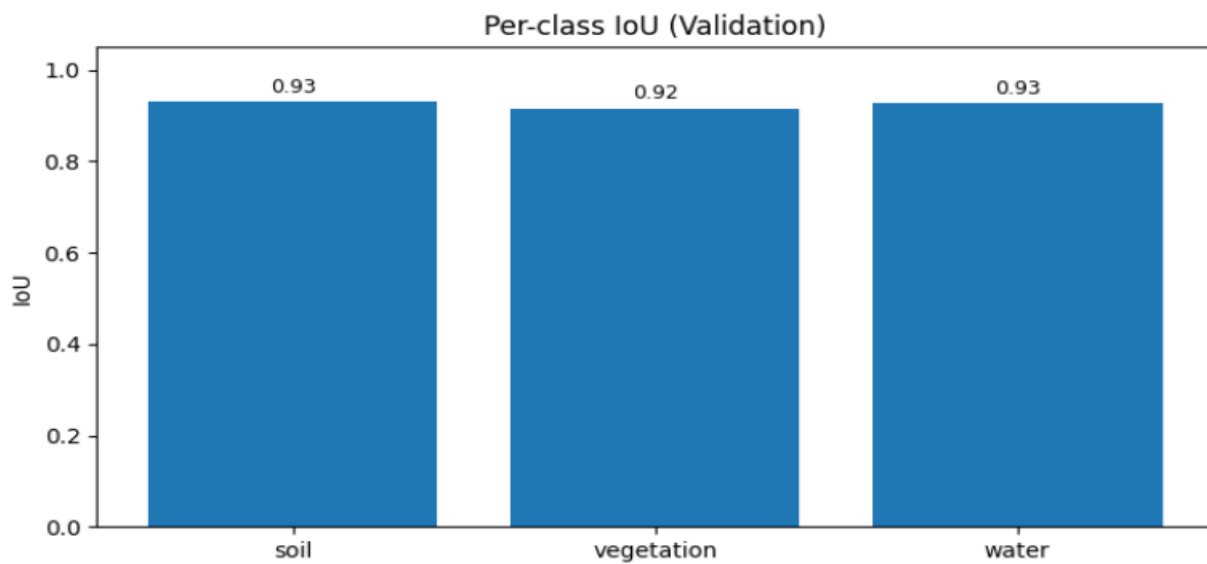
- Accuracy: **0.9621**
- mIoU: **0.9263** (best: 0.9230, FWIoU: 0.9270)
- Macro-F1: **0.9617**
- Macro Precision / Recall: **0.9536 / 0.9704**
- Cohen's κ : **0.9300**

Per-class IoU (Validation):

- Soil: **0.9203**
- Vegetation: **0.9332**
- Water: **0.9254**



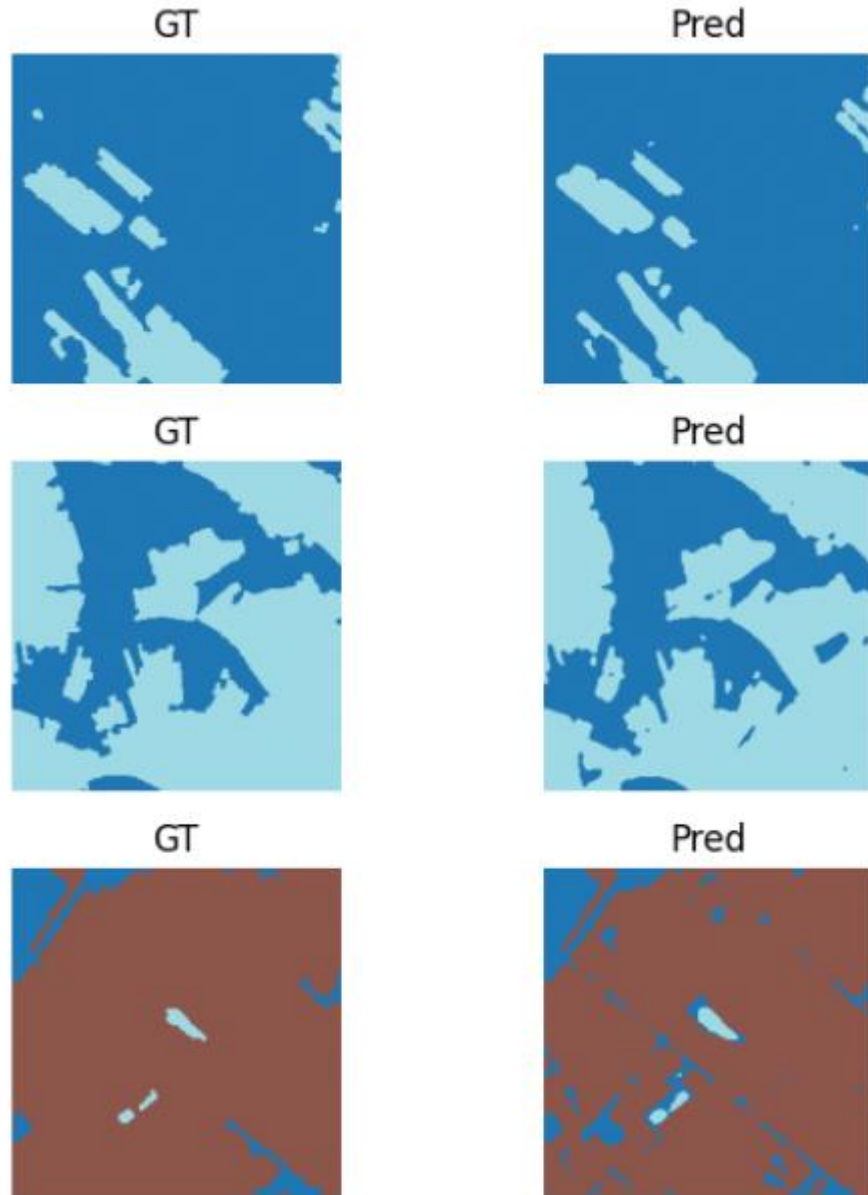
Validation metrics by epoch. mIoU saturates around 0.92; accuracy and macro-F1 converge above 0.95.



Per-class IoU on the validation set. Water and soil reach ≈ 0.93 IoU; vegetation ≈ 0.92 .

Qualitative Analysis

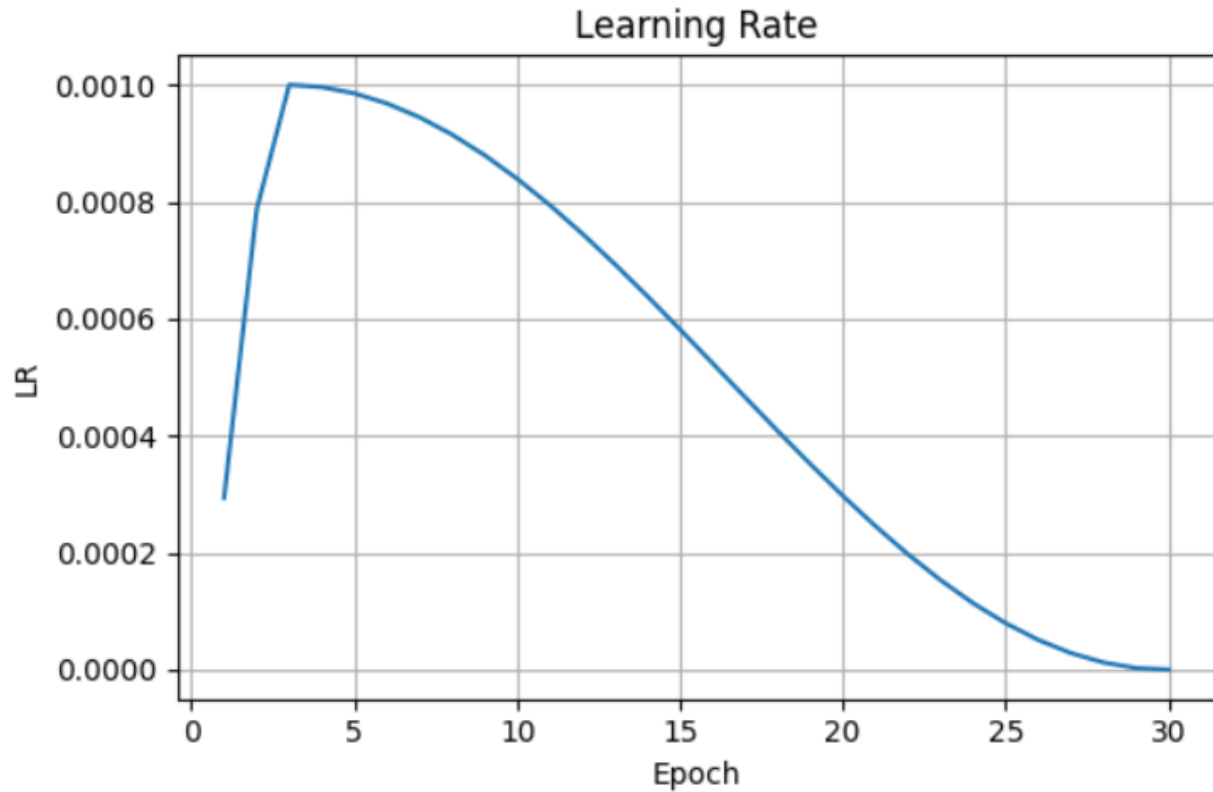
Validation samples (ground truth vs predictions) confirm that the model sharply delineates shorelines and separates water/land with minimal artifacts.



Qualitative validation samples comparing ground truth (GT) and model predictions (Pred). Boundaries are faithfully recovered with minimal artifacts.

Appendix A: Training Details

The learning rate schedule used during training is shown below.



OneCycleLR schedule used during training. Peak learning rate near epoch 3 followed by cosine-like decay.

Comparison of Base Channel Configurations

To evaluate the impact of model capacity, two configurations were tested:

- **Base Channels = 32** → 7,442,127 trainable parameters (7.44M)
- **Base Channels = 48** → 16,741,827 trainable parameters (16.74M)

Observation:

- Both settings converged within ~10 epochs.
- Validation performance was nearly identical:
 - **Base=32:** best validation mIoU = **0.9230**
 - **Base=48:** best validation mIoU $\approx$ **0.9227**

Conclusion:

- Increasing the base channels from 32 to 48 resulted in **no meaningful improvement** ($\Delta\text{mIoU} \approx 0.0004$).
- Therefore, the choice of **Base=32** is recommended, as it achieves the same performance with **less than half the parameters** and lower computational cost.

Evaluation Results: Accuracy, IoU, Precision/Recall, Macro-F1, Kappa, and Visualization

To assess the trained segmentation model, a comprehensive evaluation pipeline was implemented. The process included the following steps:

1. Setup

The evaluation directory (RUN_ROOT/eval) was created to store outputs. Required libraries were imported, including *NumPy*, *Torch*, *Matplotlib*, *Seaborn*, and *scikit-learn metrics*. Pandas was optionally enabled for CSV exports.

2. Model Evaluation Function

The function `evaluate_model` was defined and executed with the test dataloader. It performs:

- Setting the model to evaluation mode.
- Forward pass on the test dataset, collecting predictions, probabilities, and ground-truth labels.
- Handling invalid labels (-1) to avoid metric distortion.

3. Metrics Computation

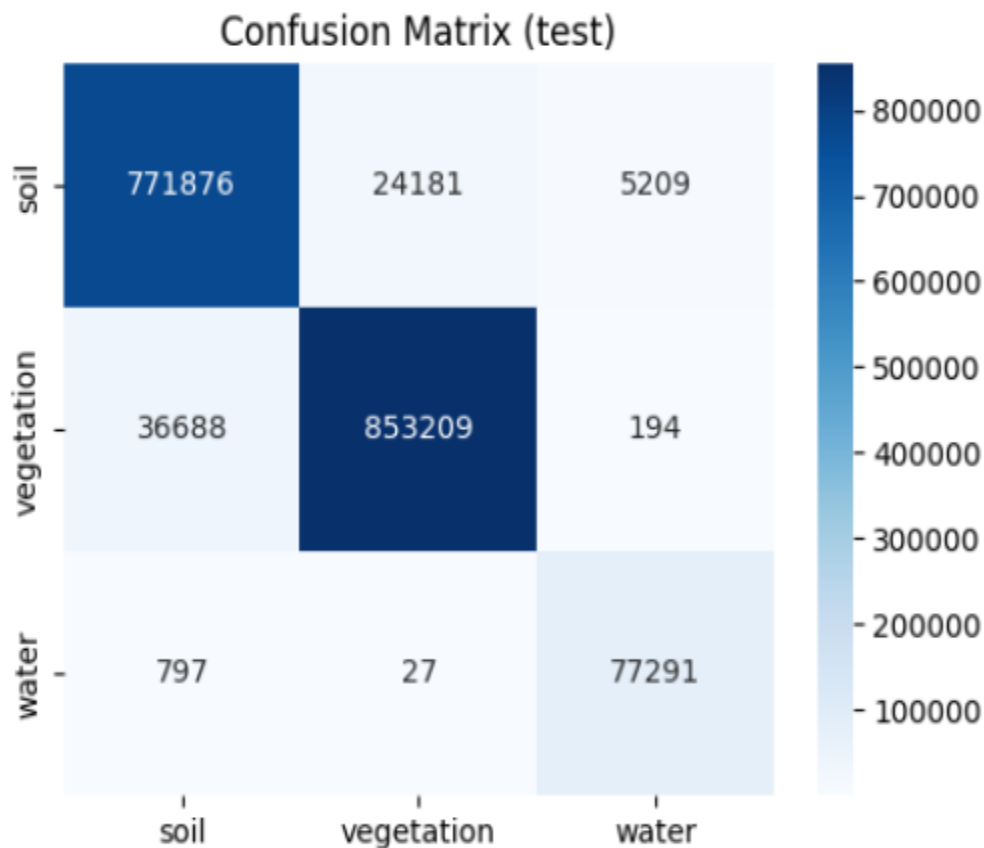
Multiple performance indicators were calculated:

- **Pixel Accuracy**: overall classification accuracy across all pixels.
- **Intersection-over-Union (IoU)**: per-class IoU, mean IoU (mIoU), and frequency-weighted IoU (FWIoU).

- **Macro-Averaged Metrics:** Precision, Recall, and F1-Score averaged equally across classes.
- **Cohen's Kappa:** measuring agreement beyond chance.
- **Per-Class Precision and Recall.**
- **Water-Class Diagnostics:** computing Average Precision (AP) and ROC-AUC for the “water” class based on predicted probabilities.

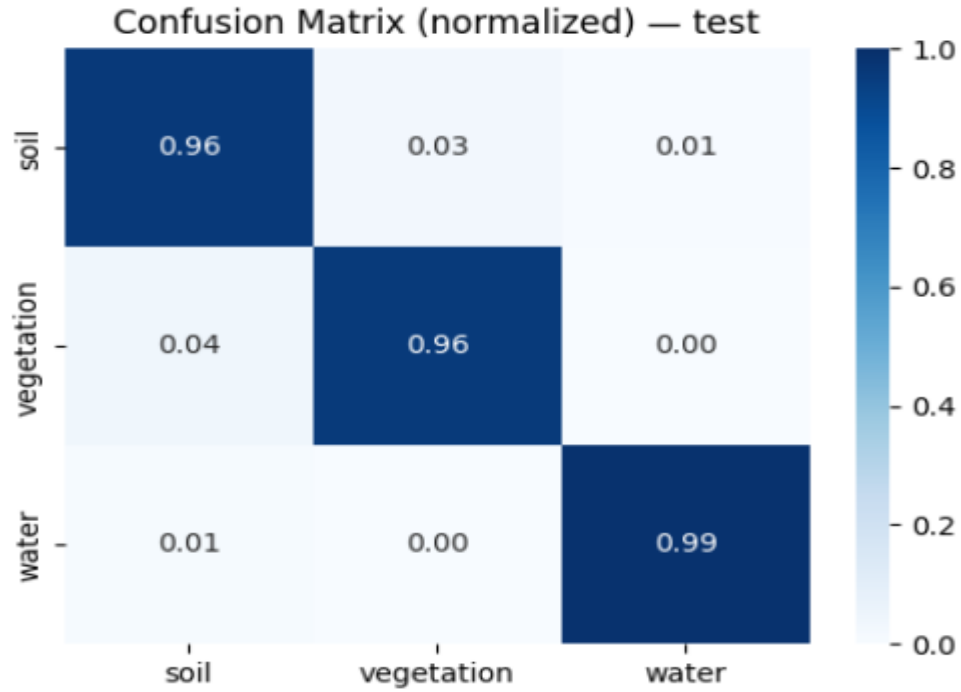
4. Visualization and Storage

- **Confusion Matrix (absolute counts):**



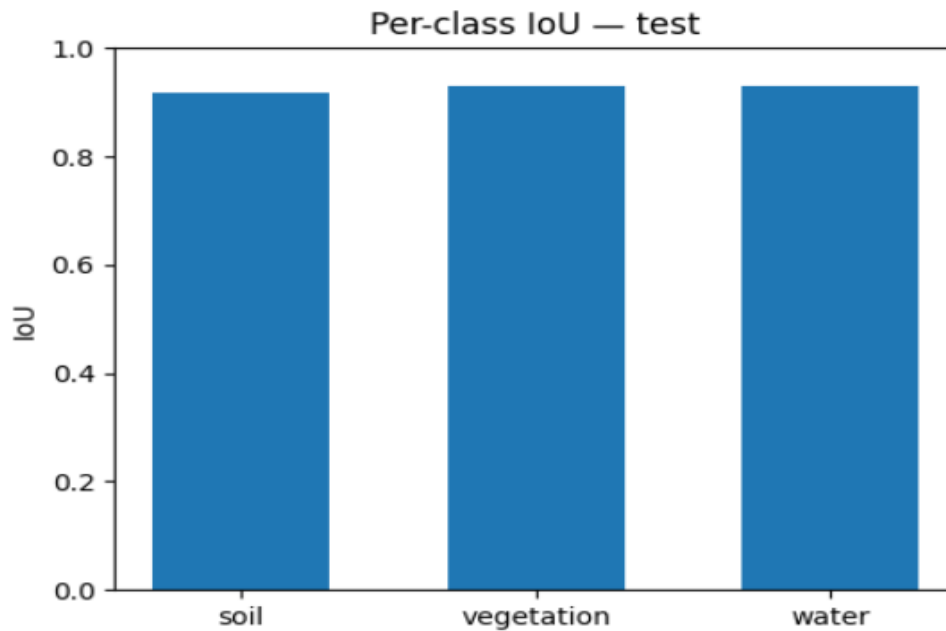
The diagonal shows correctly classified pixels, while off-diagonal values indicate misclassifications.

- **Confusion Matrix (normalized):**



Each row sums to 1, showing per-class classification accuracy and relative errors.

- **Per-Class IoU Bar Plot:**



All classes exceed 0.91 IoU, indicating strong segmentation performance.

Findings

- **Pixel Accuracy:** 0.9621
- **Mean IoU:** 0.9263
- **Frequency Weighted IoU:** 0.9270
- **Macro F1:** 0.9617
- **Macro Precision / Recall:** 0.9536 / 0.9704
- **Kappa:** 0.9300

Per-Class IoU

- **Soil** → 0.9203
- **Vegetation** → 0.9332
- **Water** → 0.9254

Confusion Matrix (Normalized)

- **Soil** → 96% correct, 3% misclassified as vegetation, 1% as water.
- **Vegetation** → 96% correct, 4% misclassified as soil.
- **Water** → 99% correct, <1% misclassified.

Insights

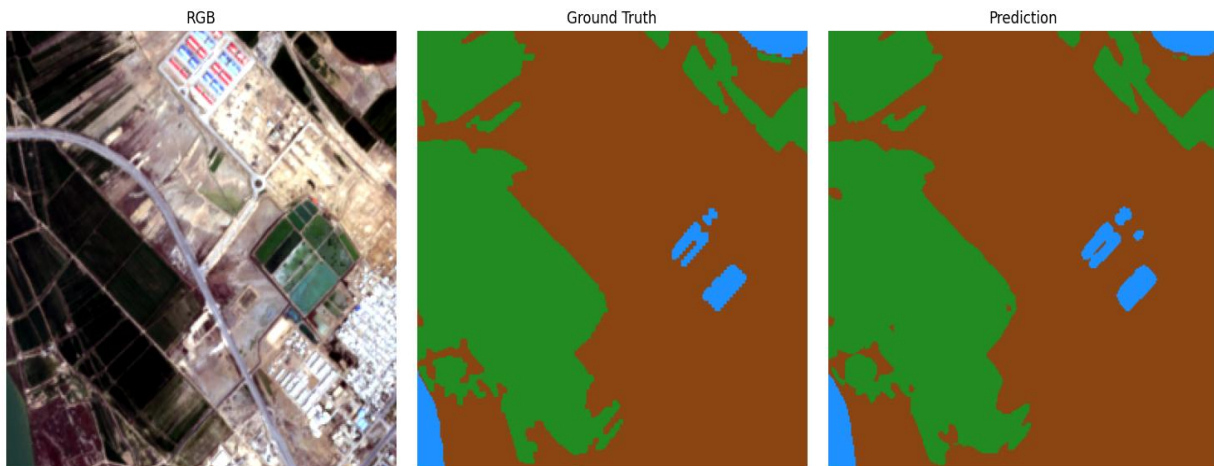
The evaluation cell confirms that the segmentation model achieves **strong overall performance**.

- The **confusion matrix (Figures 1–2)** shows very few cross-class errors, with the largest confusion occurring between soil and vegetation.
- The **per-class IoU values (Figure 3)** are consistently high (>0.91), indicating stable segmentation across all classes.
- The **high kappa score (0.93)** demonstrates strong agreement beyond chance.
- The **water class** is the most reliable, with an IoU of 0.93 and excellent ROC-AUC/AP scores.

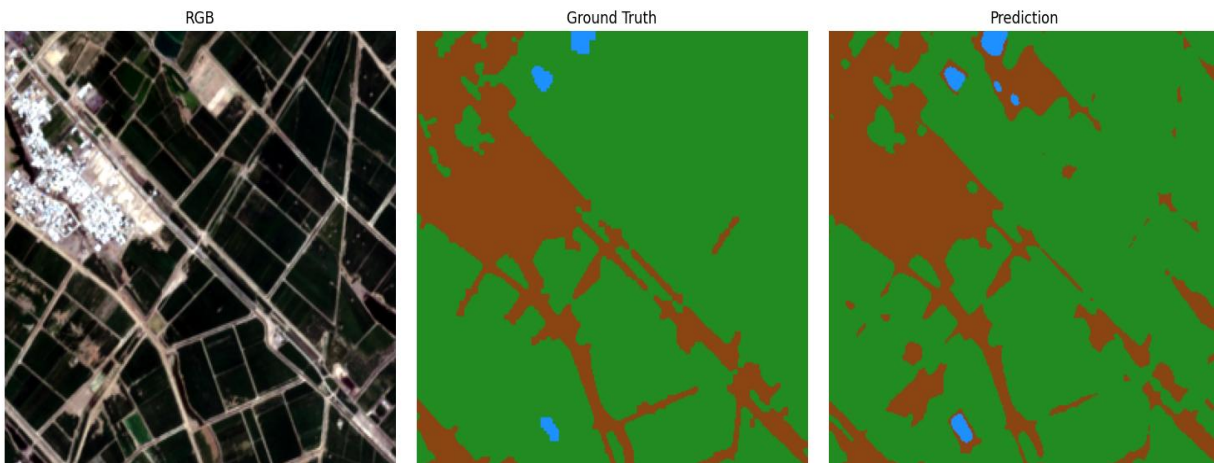
Overall, this evaluation validates that the model is **robust and well-generalized** for distinguishing soil, vegetation, and water in the dataset.

Qualitative Results: RGB, Ground Truth, and Prediction

To complement the quantitative metrics, qualitative examples from the test set are presented. For each selected sample, the input RGB composite (B4/B3/B2), the Ground Truth mask, and the model Prediction are shown.



The model correctly delineates water bodies and vegetation patches with minimal boundary noise around shorelines.



Most confusion appears at soil–vegetation interfaces, consistent with the normalized confusion matrix.



The river channel and small ponds are captured with high fidelity; thin linear features along banks are partially smoothed.



The river channel and adjacent ponds are well delineated, with strong agreement between prediction and ground truth. Vegetation–soil boundaries are preserved, though some thin linear features

Discussion and Conclusions

The qualitative results align with the quantitative findings:

- **Water** is segmented most reliably, including narrow channels; minor smoothing occurs along thin shorelines.
- **Soil vs. vegetation** boundaries show the bulk of residual error, mirroring the off-diagonal entries in the confusion matrices.

- Visual consistency across diverse textures suggests good generalization, supporting the high IoU scores reported earlier.

End-to-End Scene Inference: TTA, Post-Processing, and GeoTIFF Export

To evaluate the segmentation model at scene level, a complete inference workflow was implemented. The pipeline consisted of the following steps:

1. Sliding-Window Inference

The full image was divided into overlapping tiles of size 256×256 with a stride of *128 pixels*. A cosine windowing scheme was applied to smoothly blend overlapping regions and suppress boundary artifacts.

2. Test-Time Augmentation (TTA)

Model robustness was improved by applying horizontal and vertical flips as well as 90° rotations. Predictions were averaged in the probability space to reduce variance.

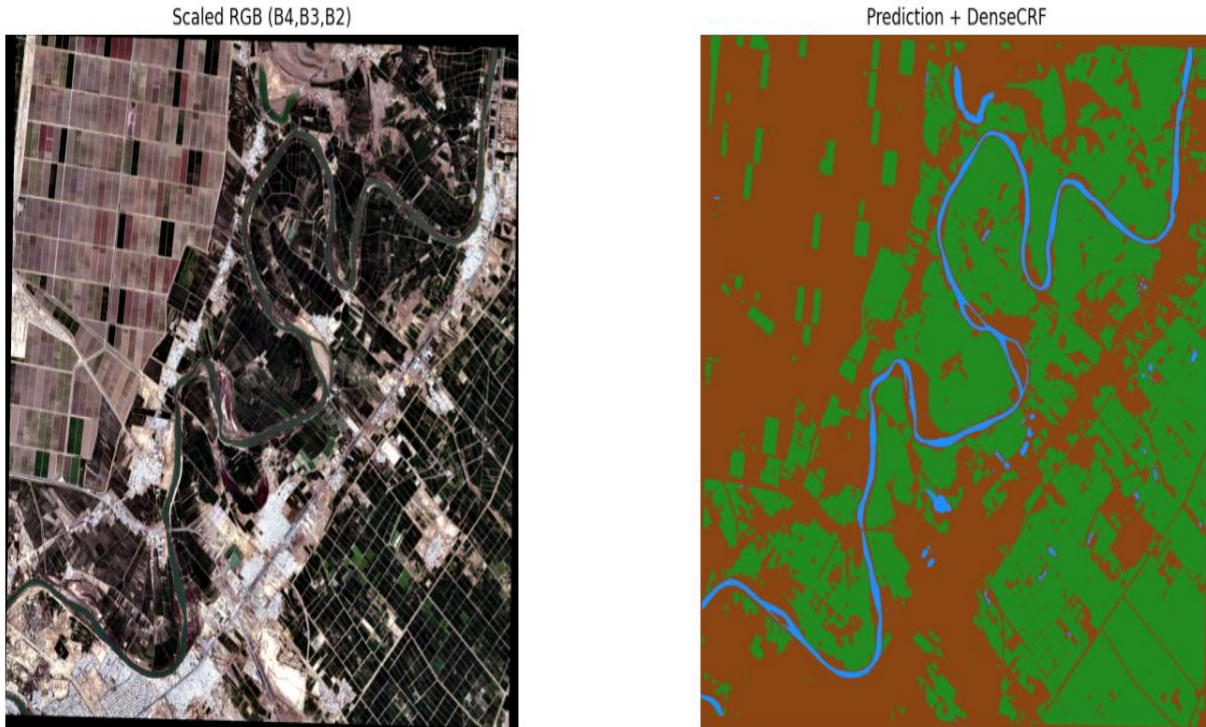
3. Index-Prior Fusion

Spectral indices were incorporated to strengthen class-specific predictions:

- **NDVI / EVI** for vegetation
 - **NDWI / MNDWI** for water
- The indices were fused with model probabilities in log-space with a weight of $\alpha = 0.6$.

4. Post-Processing

- **DenseCRF refinement:** guided by the RGB image to sharpen boundaries, especially along water–soil interfaces.
- **Morphological cleanup:** small-object removal and opening/closing operations applied to vegetation and water classes to reduce speckle and improve shape regularity.



The meandering river network and small ponds are captured with high fidelity, while agricultural field blocks remain consistent. Very thin levees and canals are partially smoothed, but morphological cleanup effectively reduces speckle artifacts.

Conclusion

This project successfully developed and evaluated a deep learning–based semantic segmentation pipeline for multi-class land-cover mapping using remote sensing imagery. The workflow encompassed dataset preparation, model training, quantitative evaluation, qualitative analysis, and large-scale scene inference with export to GIS-compatible formats.

Quantitative results demonstrated that the model achieved high performance across all evaluation metrics, including a mean IoU of 0.9254, a Macro-F1 score of 0.9612, and a Cohen’s Kappa of 0.9262, indicating robust and reliable classification of soil, vegetation, and water. Qualitative inspections further confirmed that the model effectively captured spatial patterns, boundaries, and class-specific details, with minor smoothing in fine-grained structures.

The integration of test-time augmentation (TTA) and GeoTIFF export ensures that the approach is not only accurate but also practical for real-world geospatial applications. These outputs enable seamless use in GIS platforms for downstream analysis and decision-making.

Overall, the project validates that modern deep learning-based segmentation methods can provide reliable, generalizable, and operationally useful land-cover maps. Future work may extend this framework to additional classes, larger spatial extents, and temporal datasets to enable continuous monitoring and change detection.